Assignment 1: Technical Article

Write and publish an article titled “How Excel is Used in Real-World Data Analysis” DEV.to or any other blog platform of your choice. In your article, explain what Excel is and describe how it is used in real-world scenarios. You should also include examples of features or formulas you have learned and explain how they are applied when working with data.

If you do not already have an account on DEV.to, go to the website, create an account using your email or GitHub, and complete your profile with a short bio before publishing your article. Conclude your article with a brief personal reflection on how learning Excel has changed the way you understand and interpret data, then publish it and share the link via email to datascience@luxdevhq.com while copying chepbree@luxdevhq.com by Wednesday 25th March 5pm.

Assignment 2

Data Analysis Dashboard Project

Project Title
Jumia Product Performance Dashboard: Analyzing Pricing, Discounts, and Customer Reviews

Project Objective
The objective of this project is to design and build an interactive Excel dashboard that analyzes the performance of products listed on Jumia. Students will explore how pricing, discounts, reviews, and ratings influence product performance and customer engagement. The final dashboard should support data-driven decision-making in an e-commerce context.

Dataset Overview
The dataset contains product-level data with the following columns:
Product – Name of the product
Current Price – Current selling price in Kenyan Shillings (KSh)
Old Price – Original price before discount in Kenyan Shillings (KSh)
Discount – Discount percentage applied to the product
Reviews – Number of customer reviews
Rating – Average customer rating out of 5

Data Cleaning and Preparation
• Check for and handle missing values, especially in the Reviews and Rating columns
• Identify and remove duplicate product entries
• Convert price columns into numeric format by removing "KSh" (use split text to columns under Data tab), commas, and text or use Find and Replace (Ctrl + H)
• Ensure the Discount column is numeric and properly formatted as a percentage (use Find and Replace, LEFT/RIGHT functions, or Text to Columns)
• Convert the Rating column from text format (e.g., "4.5 out of 5") into numeric values
• Convert negative reviews to positive

Data Enrichment
Create the following additional columns:
• Discount Amount (KSh): Old Price minus Current Price
• Rating Category:
  Poor for ratings below 3
  Average for ratings between 3 and 4.4
  Excellent for ratings of 4.5 and above
• Discount Category:
  Low Discount for discounts below 20%
  Medium Discount for discounts between 20% and 40%
  High Discount for discounts above 40%

Data Analysis

Descriptive Analysis
• What is the average current price, old price, discount percentage, and rating?
• Which product is the most expensive and which is the least expensive?
• What is the average rating and discount by discount category?
• How are products distributed across rating categories?

Trend and Relationship Analysis
• Analyze the relationship between discount percentage and number of reviews
• Analyze the relationship between rating and number of reviews
• Determine whether higher discounts lead to increased customer engagement
• Determine whether higher-rated products tend to have more reviews

Product Performance Analysis
• Identify the top 10 products with the highest discounts
• Identify the top 10 products with the most reviews
• Identify the top 5 highest-rated and bottom 5 lowest-rated products
• Compare high-discount and low-discount products based on average rating and number of reviews

Dashboard Design

Dashboard Requirements
Create a single interactive Excel dashboard containing the following sections:

Overview
• Total number of products
• Average rating
• Average discount percentage
• Total number of reviews

Product Performance
• Top products by rating
• Top products by number of reviews
• Top products by discount percentage

Trend Analysis
• Visualizations showing discount percentage versus reviews
• Visualizations showing rating versus reviews

Product Categories
• Breakdown of products by rating category
• Breakdown of products by discount category

Visualization Guidelines

- Use Pivot Tables as the primary data source
- Use appropriate charts such as bar charts, column charts, pie or donut charts, and scatter plots
- Apply conditional formatting to highlight high discounts and low ratings
- Include slicers for rating category, discount category, and price range where applicable

Submission

Submit your completed Excel dashboard by Wednesday 25th March 5pm by sharing it via email to datascience@luxdevhq.com and copy chepbree@luxdevhq.com.